

b. Find the Nyquist rate of sampling for

i.  $x(t) = \text{sinc}^2(50t)$

ii.  $\text{sinc}(100t)\sin(200\pi t)$

Also write the expression for the signal reconstructed from the samples.

**Paper Code: ECC 08**

Time: 3.00 Hours

**Subject: Signals and Systems**

Maximum Marks: 50

*Note: Attempt ANY FIVE questions. All questions carry equal marks. Assume suitable data if necessary.*

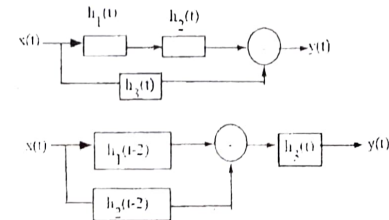
### Q.1

- a. (i) Given that  $\text{sinc}(at) * \text{sinc}(bt) = A\text{sinc}(Bt)$ ,  $a > b$ . Find the constants A, B.  
 (ii) If  $e^{-at}u(t) * e^{-at}u(t) = f(t)e^{-at}$  find  $f(t)$ .

b. State and prove the properties of ROC of Z transform.

### Q.2

- a. If  $h_1(t) = u(t+2)$ ,  $h_2(t) = u(t-2)$ ,  $h_3(t) = e^{-at}u(t)$ . Find out if the following system shown in figure below are stable, causal.



- b. An LTI system is described by the difference equation  $y[n] = y[n-1] + y[n-2] + x[n-1]$ . Find the system function and plot the poles and zeros. Also find the impulse response corresponding to a stable system and an unstable systems that satisfies this difference equation.